Airbnb Prices in New York City

Pragyan Kadel, Meenakshy Manju, Aby Joe Jose

1 Clarkson University, Potsdam, USA

2 Department of Applied Data Science, Clarkson University, Potsdam, NY 13676, USA

Abstract Airbnb in New York City plays a significant role in shaping urban tourism, housing availability, and the local economy. It offers flexible, often affordable accommodation options for travelers while generating income for residents. Studying its impact is crucial for understanding short-term rental dynamics, policy implications, and how it influences neighborhood demographics, affordability, and city planning decisions. This project investigates the factors affecting Airbnb prices in New York City by analyzing the impact of neighborhood, room type, and customer feedback. Using statistical and machine learning methods, the project aims to predict rental prices whether it is above or below \$500 per night based on location, type of accommodation, and customer reviews.

Keywords Airbnb, price prediction, New York City, Machine Learning, Logistic Regression, Decision Tree, Random Forest

1 Introduction

Founded in 2008, Airbnb is a San Francisco,California-based online booking business that has grown to be a desirable alternative in the hospitality sector. Since then, it has expanded and dispersed globally. It currently operates in 81,000 cities across more than 220 countries, has millions of listings, and generates billions of dollars in revenue. With the rapid growth of the sharing economy, platforms like Airbnb have transformed how people travel and find accommodations. In a city as dynamic and diverse as New York City, Airbnb listings vary widely in terms of location, room type, and price. Understanding what drives these price differences is crucial for both hosts and guests. Unlike conventional hotels, Airbnb lets hosts determine their own rates based on their personal experiences. For new landlords, choosing the ideal pricing without losing clients is a difficult task. It's always helpful to know whether the current price is fair and whether now is a good time to make a reservation, even though customers may compare similar pricing. In summary, predicting housing prices remains a widely researched and practically significant challenge. This study aims to evaluate and compare the accuracy of various machine learning models in forecasting rental prices. The process begins with careful selection and pre-processing of the dataset, including handling of missing values to ensure data quality. Following this, data visualization techniques are employed to explore and understand the relationships between key influencing factors and rental prices. Subsequently, multiple machine learning algorithms are implemented to perform price prediction, and their performance is assessed using metrics such as accuracy, precision and recall. The results provide insights into the relative effectiveness of each model and highlight the most suitable approach for this dataset. These findings can serve as a valuable reference for real estate analysts and decision-makers seeking data-driven pricing strategies.

2 Methods

2.1 Source of data

This project utilizes a dataset comprising 42,921 Airbnb listings from the years 2011 to 2023, sourced from the publicly available Airbnb open dataset on Kaggle. The dataset initially contains 18 columns; however, certain fields such as id and host_id are excluded from the analysis, as they do not serve as meaningful predictors for rental pricing. These identifiers do not provide explanatory value for the modeling process and are therefore removed. Additionally, the license column, which contains entirely null values and holds no analytical relevance, is also omitted during the data pre-processing phase to maintain dataset integrity and efficiency.

2.2 Variables

2.2.1 Dependent Variable

price - Daily price in local currency (US Dollars)

2.2.2 Independent Variables

id - Airbnb's unique identifier for the listing
name - Name of the listing
host_id - Airbnb's unique identifier for the host/user
host_name - Name of the host. Usually just the first name(s)
neighborhood_group - The neighborhood group as geocoded using the latitude and longitude against neighborhoods as defined by open or public digital shapefiles
neighborhood - The neighborhood as geocoded using the latitude and longitude against neighborhoods as defined by open or public digital shapefiles
latitude - Uses the World Geodetic System (WGS84) projection for latitude and longitude
longitude - Uses the World Geodetic System (WGS84) projection for latitude and longitude
room_type - Indicates the four types of room. Private room, Shared room, Hotel room, Entire room/Apt
minimum_nights - minimum number of night stay for the listing
number_of_reviews - The number of reviews the listing has
last_review - The date of the last/newest review
reviews_per_month - Number of review per month
calculated_host_listings_count - The number of listings the host has in the current scrape, in the city/region geography
availability_365 - The availability of the listing x days in the future as determined by the calendar
number_of_reviews_ltm - Number of reviews of the last 12 months

2.3 Data Processing

The columns, name and host_name contain 12 and 5 missing values respectively. These null entries were replaced with the placeholder value "unknown". Additionally, 10,304 values were missing from both the last_review and reviews_per_month columns. To address this, the SimpleImputer was employed: missing values in last_review were imputed with 0, while missing values in reviews_per_month were replaced using the median of the available data.

The objective of the model is to predict whether a hotel's price per night falls under \$500 or above \$500. During initial exploration, it was observed that prices exceeding \$500 constituted a minority class in the dataset. To address this class imbalance and enhance the model's ability to accurately learn from both classes, the Synthetic Minority Oversampling Technique (SMOTE) was applied. Specifically, SMOTE was used to increase the number of instances in the minority class by 30%, creating a more balanced dataset for model training.

2.4 **Feature Significance**

2.4.1 Normality Test

Shapiro-Wilk test is used to assess the normality of each numeric column. Based on the p-value, if p is greater than 0.05, the column appears normally distributed (fail to reject the null hypothesis of normality) or else if p is less than or equal to 0.05, the column is not normally distributed (reject the null hypothesis). After analyzing it was found that the numeric columns and the target variables are not normally distributed.

2.4.2 Kruskal-Wallis Test

The Kruskal–Wallis H-test was employed to evaluate the statistical significance of individual categorical features in relation to Airbnb listing prices. Given that the dataset did not meet the assumptions required for parametric tests, particularly the assumption of normality, a non-parametric approach was deemed appropriate. The Kruskal–Wallis test is well-suited for comparing the medians of multiple independent groups when the data distribution is non-normal.

The features such as neighborhood_group and room_type were independently assessed to determine whether significant differences existed in median listing prices across its respective categories. The test yielded both H-statistics and associated p-values. The two features demonstrated particularly strong significance, reflected in high Kruskal–Wallis H-statistics and p-values less than 0.05. These findings highlight substantial variations in listing prices based on these categorical groupings.

2.4.3 Feature Selection

In order to reduce dimensionality and evaluate the contribution of each feature in the dataset, Principal Component Analysis (PCA) was performed. The objective of this analysis was to determine the number of principal components needed to capture a significant proportion of the dataset's variance while minimizing information loss.

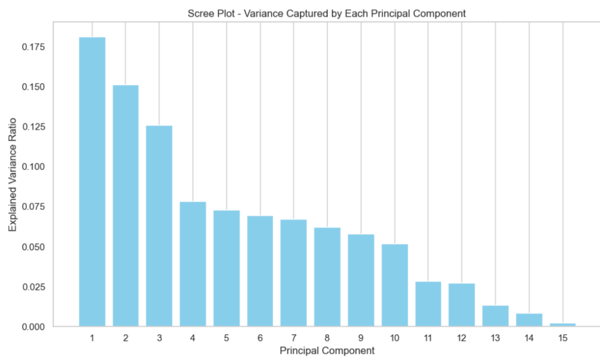

Fig. 1 Scree Plot

The scree plot (Fig 1) represents the variance explained by each principal component individually. It can be observed that the first few components capture a substantial amount of variance, with a sharp decline after the first three components, followed by a gradual decrease across the subsequent components. The first principal component explains approximately 18% of the total variance, while the second and third components explain about 15% and 13% respectively. After the tenth component, the variance contribution of each additional component drops significantly, indicating diminishing returns in information gain.

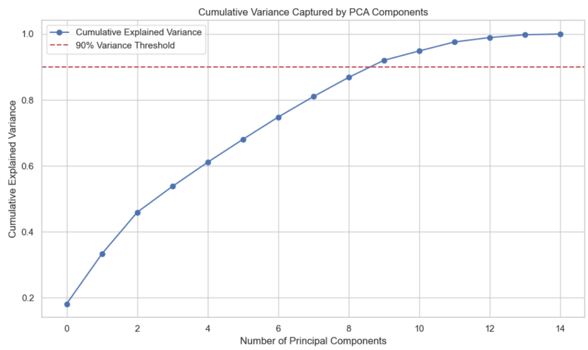

Fig. 2 Cumulative Variance Captured by PCA Components

The cumulative variance graph provides further insights by illustrating the total variance captured as additional components are considered. It shows that approximately 90% of the total variance is retained by the first nine principal components. Extending to the tenth principal component allows the cumulative explained variance to reach approximately 94%, thereby capturing a larger portion of the dataset's variability with minimal additional complexity.

Based on these findings, we decided to retain the first ten principal components for further analysis. This selection captures approximately 94% of the total variance, ensuring that the majority of the dataset's information is preserved while reducing the dimensionality of the data. Using ten principal components strikes a practical balance between maintaining model performance and improving computational efficiency, while also mitigating the risk of overfitting and reducing noise.

3 Model Building

For the Airbnb price prediction, Logistic Regression, Decision Tree and Random Forest models were selected due to their effectiveness in handling classification problems and their interpretability.

3.1 Logistic Regression

Logistic Regression was employed to classify Airbnb listings based on whether their price per night was less than or greater than \$500. The model achieved an overall accuracy of 79%, demonstrating reasonable performance on the dataset. It showed strong predictive ability for listings priced below \$500, achieving a precision of 0.82 and a recall of 0.94. These metrics suggest that the model was highly effective in correctly identifying the majority class, ensuring that most lower-priced listings were classified accurately. This strong performance indicates that the model captures the underlying patterns associated with more affordable listings reasonably well.

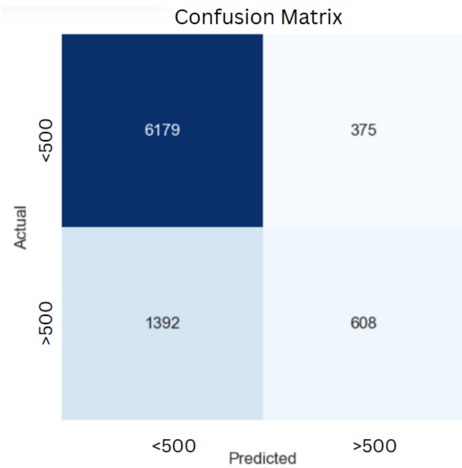

Fig. 3 Confusion Matrix of Logistic Regression

However, the model's performance in predicting listings priced above \$500 was significantly weaker. The precision for this minority class dropped to 0.62, and the recall decreased to 0.30, indicating a considerable number of false negatives where expensive listings were misclassified as cheaper ones. Analysis of the confusion matrix confirmed that a large portion of listings priced above \$500 were incorrectly predicted as being under the threshold. This imbalance in performance highlights a bias toward the majority class, a common issue when the training data contains a significant class imbalance. Addressing this bias through additional techniques, such as more extensive oversampling or the use of alternative classification algorithms, could further improve the model's predictive capabilities for higher-priced listings.

3.2 Decision Tree

A Decision Tree model was constructed to classify Airbnb listings based on whether their price per night was less than or greater than \$500. The model demonstrated strong overall performance, achieving an accuracy of 90%. It performed consistently well across both classes, with the less than \$500 class achieving a precision of 0.94 and a recall of 0.93. For listings priced above \$500, the model also performed reliably, obtaining a precision of 0.78 and a recall of 0.79. This balanced performance indicates that the Decision Tree was effective in minimizing misclassifications for both the majority and minority classes, outperforming the earlier Logistic Regression model.

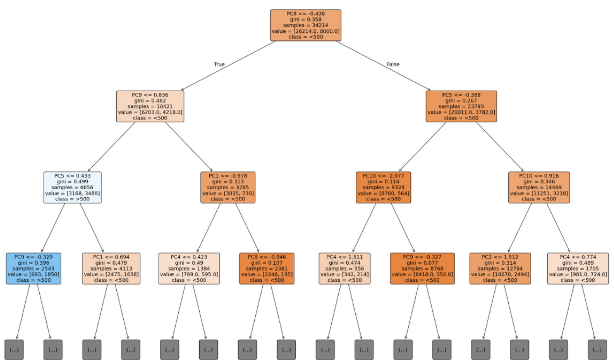

Fig. 4 Tree Map

Further analysis of the confusion matrix confirmed that the Decision Tree significantly reduced classification errors compared to previous models. Additionally, one of the key advantages of using a Decision Tree is its interpretability. The visual representation of the tree structure clearly illustrates the decision paths based on important features, providing valuable insight into how the model arrives at its predictions. This interpretability not only aids in understanding model behavior but also supports better communication of results to stakeholders.

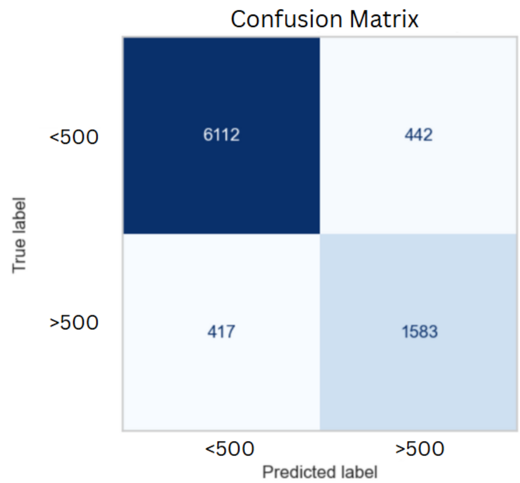

Fig. 5 Confusion Matrix of Decision Tree

Figure 4 presents a visualization of the Decision Tree, displaying only 4 levels of the tree structure. This partial representation is provided for clarity and ease of understanding, as the original Decision Tree contains 17 levels. The decision tree's full depth and complexity are not shown here due to space limitations, but the selected 4 levels illustrate the key decision-making process.

3.3 Random Forest

The Random Forest model exhibited exceptional performance, achieving an overall accuracy of 94%. It demonstrated high precision and recall across both classes, with particularly strong results for the minority class (> \$500). In this class, the model achieved a precision of 0.89 and a recall of 0.81, indicating its ability to correctly identify listings in this higher price range. The model's ability to effectively distinguish between lower and higher pricing segments suggests its robustness in handling varying class distributions, making it a versatile tool for diverse data types. This ability to maintain consistent performance across both common and rare classes highlights the model's effectiveness in real-world scenarios where class imbalance is often present. Furthermore, Random Forest's ability to generalize well across different datasets ensures its adaptability to various business applications. Its flexibility makes it suitable for a wide range of industries, from e-commerce to real estate.

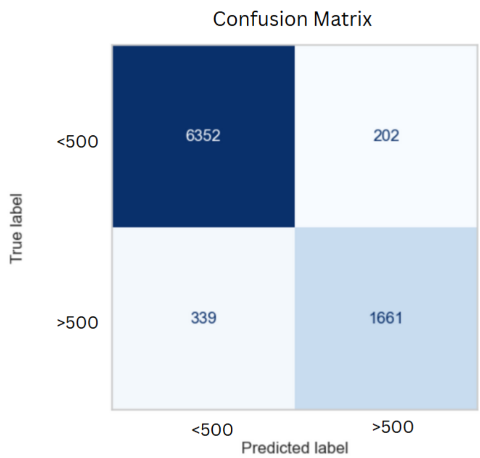

Fig. 6 Confusion Matrix of Random Forest

The confusion matrix further supports the model's reliability, showing very few misclassifications. This outcome underscores the model's dependability in predicting both lower and higher listing prices. Notably, Random Forest's performance in handling class imbalance is a significant strength, making it a suitable and trustworthy solution for predicting listing prices across different pricing segments. Additionally, the model's stability across different data sets and conditions enhances its potential for widespread application in real-world scenarios.

3.4 Feature Importance

The Decision Tree model emphasized PC8 as the most important principal component, followed closely by PC9, PC5, and PC2. These components carried the most predictive weight, indicating that the underlying features grouped into these principal components had a significant impact on determining the price category. The importance scores dropped gradually, suggesting that only a few components strongly influenced the model's predictions.

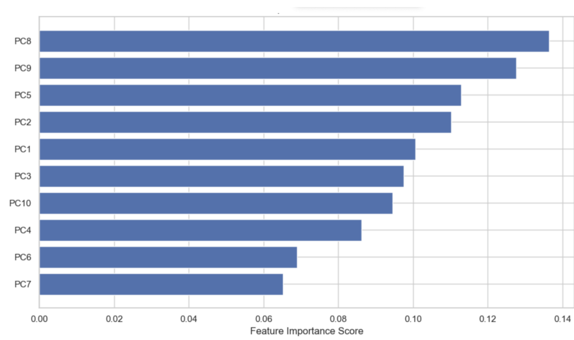

Fig. 7 Feature Importance of Decision Tree

In contrast, the Random Forest model also ranked PC8 and PC9 as the most influential components, but it displayed a more balanced distribution of importance across additional components such as PC1, PC3, and PC5. This broader spread reflects the ensemble nature of Random Forest, which leverages multiple decision trees to consider complex patterns and interactions between features more effectively.

Both models consistently identified PC8 and PC9 as key drivers in predicting listing price categories. This consistency reinforces their reliability as significant features. However, while the Decision Tree model concentrated its importance on a smaller subset of components, the Random Forest model distributed importance more evenly, offering better generalization and robustness in handling diverse data patterns.

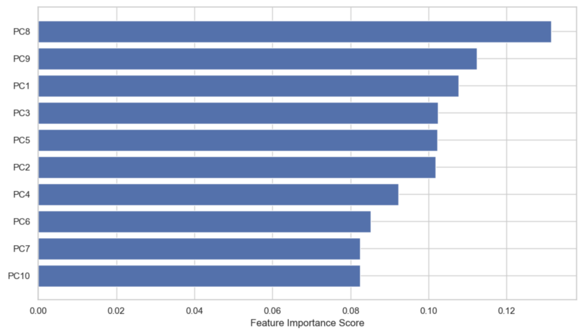

Fig. 8 Feature Importance of Random Forest

Original Features	PC8 Coefficient
Minimum Nights	0.76
Neighborhood Group Staten Island	0.51
Availability 365	-0.19
Neighborhood Group Queens	0.19
Number of Reviews	0.17
Neighborhood Group Manhattan	0.17
Neighborhood Group Bronx	-0.08
Number of Reviews ltm	0.08
Latitude	-0.05
Neighborhood Group Brooklyn	-0.05
Reviews per month	0.04
Room type Entire home/Apt	-0.01
Room type Hotel room	0.01
Longitude	0.00

Table 1 PC8 Contributors

Original Features	PC9 Coefficient
Room type Entire home/Apt	0.71
Room type Hotel room	0.57
Neighborhood Group Staten Island	0.30
Minimum Nights	-0.16
Neighborhood Group Manhattan	-0.15
Calculated host listings count	0.14
Latitude	0.08
Longitude	-0.08
Neighborhood Group Bronx	0.07
Number of Reviews	0.03
Availability 365	0.02
Number of Reviews ltm	0.02
Neighborhood Group Queens	0.02
Reviews per month	0.01
Neighborhood Group Brooklyn	0.00

Table 2 PC9 Contributors

An analysis of the principal components PC8 and PC9, which were consistently ranked as the most influential in both Decision Tree and Random Forest models, reveals strong contributions from features related to room type, neighborhood, and listing activity. PC8 was heavily influenced by variables such as Minimum Nights, Neighborhood Group Staten Island, and Number of Reviews, indicating that booking behavior and location significantly drive this component. Meanwhile, PC9 placed strong weight on Room Type Entire Home/Apt, Room Type Hotel Room, and Neighborhood Group Staten Island, emphasizing the importance of accommodation type in shaping price categories. The convergence of room type and geographic variables across both PCs highlights their critical role in price prediction, underscoring their relevance in both tree-based models.

4 Results

Table 5 shows the accuracy of different models in predicting prices. In this project, the model is judged by comparing the accuracy value.The modeling process indicated that the Random Forest outperformed Logistic Regression and Decision Tree, providing a higher accuracy of 94% and better capturing feature complexities in predicting Airbnb prices.

Principal Component Analysis effectively reduced feature dimensionality, retaining 95 percent variance with 10 components, facilitating efficient modeling without substantial information loss. Kruskal-Wallis tests affirmed feature significance, highlighting critical predictors like room type and location. Future analyses could explore additional advanced algorithms, include more detailed feature engineering, and assess model performance on similar datasets from different geographical regions to validate generalization.

Model	> $500		
	Precision	**Recall**	**F1**
Logistic Regression	0.62	0.30	0.41
Decision Tree	0.78	0.79	0.79
Random Forest	0.89	0.81	0.84

Table 3 Model Performance Metrics for > $500 class

Model	< $500		
	Precision	**Recall**	**F1**
Logistic Regression	0.82	0.94	0.87
Decision Tree	0.94	0.93	0.93
Random Forest	0.94	0.97	0.96

Table 4 Model Performance Metrics for < $500 class

Model	Accuracy
Logistic Regression	79%
Decision Tree	90%
Random Forest	94%

Table 5 Comparison of Model Accuracy

5 Conclusion

The model successfully developed predictive models for Airbnb pricing, emphasizing effective data pre-processing, statistical testing, and dimensionality reduction using PCA. The Random Forest model emerged superior with a notable accuracy, further optimized through grid search, highlighting complex relationships within features. Significant predictors identified through Kruskal-Wallis provided insights, particularly regarding location and room type. Overall, these models ensure fairly accurate price prediction, offering valuable implications for hosts, customers, and strategic pricing decisions on rental platforms.

Acknowledgements We sincerely thank our instructor for their guidance throughout this project. We also acknowledge Airbnb for providing the NYC Open Data used in our analysis and appreciate the collaborative efforts of all team members.

References

1. James G, Witten D, Hastie T, Tibshirani R. An Introduction to Statistical Learning: With Applications in R. Springer, 2013
2. Breiman L. Random forests. Machine Learning, 2001, 45(1): 5–32
3. Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, Duchesnay É. Scikit-learn: Machine learning in python. Journal of Machine Learning Research, 2011, 12: 2825–2830
4. Jolliffe I T, Cadima J. Principal component analysis: a review and recent developments. Philosophical Transactions of the Royal Society A, 2016, 374(2065): 20150202
5. Airbnb . Airbnb nyc open data, 2021. https://www.airbnb.com/datasets/new-york-city
6. Kuhn M, Johnson K. Applied Predictive Modeling. Springer, 2013
7. Field A, Miles J, Field Z. Discovering Statistics Using R. SAGE Publications, 2012
8. Montgomery D C, Peck E A, Vining G G. Introduction to Linear Regression Analysis. Wiley, 2021
9. Liaw A, Wiener M. Classification and regression by randomforest. R News, 2002, 2(3): 18–22
10. Kassambara A. Practical Guide to Principal Component Methods in R. STHDA, 2017
11. Bruce P, Bruce A, Gedeck P. Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python. O'Reilly Media, 2020
12. Han J, Kamber M, Pei J. Data Mining: Concepts and Techniques. Morgan Kaufmann Publishers, 2011